

DDR3 for Core i5

Kingston starts shipping DDR3 kits for Core i5 processors

AMD shows off DX11

AMD demonstrates DirectX 11 hardware running six LCD screens at once

Motherboard test

discovery. In general most integrated graphics boards are based on entry-level chipsets. However, AMD's latest 790GX/FX chipsets are high-end offerings but they also retain an integrated graphics solution. Therefore we put these boards into the high-end segment, where they were tested with a discrete graphics card. Core i7 boards were tested separately, for their performance is incomparable to other platforms – obviously on account of the faster processors they support.

Intel Core i7 motherboards

Core i7-based motherboards haven't been affordable and the cost of the board along with the price of DDR3 memory has put many upgraders off. Prices have fallen, and there are many options available as you are about to see; although they are all based on a single Intel X58 chipset. These boards do not have a memory controller, since the Core i7 processor now integrates the controller on its die.

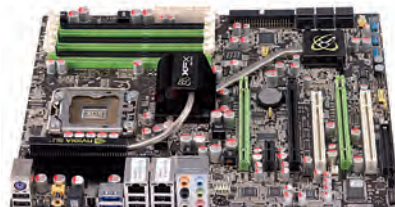
Board design and features

While the X58 boards are all extremely high-end and extremely costly, there's a trend to be followed – something every motherboard manufacturer follows. The mATX form factor means slightly less expensive motherboards and it's something that will also find its way into compact PCs and maybe even high-performance HTPC systems. MSI's X58M is one of the few mATX format motherboards we tested.

The BIOS on the X58M is thorough with MSI's distinctive overclocking utility called the Cell menu. It allows for accurate overclocking and overvolting. Another interesting menu is the

Green power menu that allows more efficient use of power using the CPU phase control. There's also a neat little reading that displays power efficiency. BIOS updating is also simple using a flash drive. Just going by features and how well it's designed, it's a pretty complete a motherboard that's not overly expensive at Rs. 10,500.

XFX and ZOTAC, both



XFX X58i

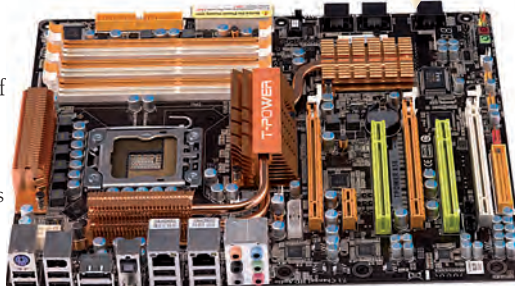
NVIDIA only vendors prior to this have built boards based on the X58 chipset. XFX calls its board the XFX X58i. The board is designed and built really well.

Features and BIOS layout are good. In terms of board layout, it's similar to the MSI X58M board with a similar level of overclocking features and thought put into it. It's a full ATX board and has a cleanly routed set of heat pipes running from one end of the board to the other. Temperatures are lower than the MSI board. There is plenty of space between PCI slots and PCIe slots which mean you won't have any troubles with plugging in sound cards with a large graphics card coming in the way.

There is an LED poster and dedicated power switches on the board. Although a great board, XFX has persisted with the ageing ALC888 sound solution – a pity. SATA ports are placed along the side like many other boards, but slightly away from the edge of the

board which is inconvenient. The PCIe x1 slot is right next to the first PCIe x16 slot, so no chance of setting up sound cards there. You'll have to use one of the other two PCIe slots. Two PCI slots seem like overkill for a board of this generation. One of the smaller complaints is the XFX site design and getting BIOS updates can be a little annoying.

ZOTAC's X58, is, in comparison to the others, not as impressive. One problem is the lack of basic necessary overclocking and memory tweaking options. One of the biggest flaws was the lack of the memory timing settings. If you have low-latency, fast memory you'll be sorely disappointed. There is no manual option for preventing CPU throttling down due to power saving features like EIST and stepping options – all these are absent in the BIOS.



Biostar Tpower X58

The benchmarks especially the memory intensive ones suffered the most. The cooling solution used is extremely basic, but doesn't get very hot with an air-conditioning unit running. In general, it feels very cheap. For example, the Northbridge heatsinks shift a bit if you touch it. Not all is bad though – the board is lined with solid capacitors along vital region. Additionally, in all fairness it is one of the less expensive X58 boards around.

MSI had another board on this platform, the X58 Pro. Unlike the X58M, the X58 Pro is an ATX board.

The larger board means that there is more space between the PCIe x16 slots, but not enough to plug in a second card in the adjoining slot. The first x1 slot lies right under the first x16 slot. The second x1 lies above the x16 slot. Longer sound cards might still have some problem fitting in there because of the high heat sink that lies very close in the way.

For those overclockers or those who might run open rigs, there are dedicated buttons for power, reset and also clearing the CMOS. Right next to it are the DIP switches that allow quick switching and overriding of the base clock speed. The BIOS is also quite elaborate and will be good enough for anyone planning on overclocking with cooling of the board itself the only thing to worry about.

BioStar's TPower X58 was next. There are LEDs for each of the 6 SATA ports on the side. Next to those slots is the poster and the lit power and reset buttons. Cooling is taken care of by a combination of heatsinks and heatpipes. The overclocking menus are well sorted out and the increments for voltages are small and good enough for anyone who might want to overclock this board.

Gigabyte's EX58-UD3R is an ATX form factor board with only four memory slots. This is a rarity as most boards have six. Like many other boards, the adjoining slot to the first x16 slot and in this case the PCIe x4 slot will be covered by the graphics card. There are two x1 slots above the first x16 slots, one of which is blocked by the gigantic northbridge heatsink. The second slot is unobstructed. The cooling solution is similar



DFI UT X58 Lanparty

If you want to know exactly what processor is best for your needs, check out the September issue for a detailed processor comparison test. When you're done with that, you will know which motherboard category you should look for